Anjan Roy

Software Engineer @ Avail Project; PhD in Applied Cryptography @ Radboud University, The Netherlands

Objective :

To work in an environment that provides high standards of intellect, professionalism and moral ethics; solve problems that helps us in understanding the mysteries of the nature; be part of a team that works towards growth of society and enhance my skills while adopting newer methodologies.

Email : hello@itzmeanjan.in
Website : https://itzmeanjan.in
Github: https://github.com/itzmeanjan

Gender : Male
Nationality : Indian

Experience :

Engineering @ Avail Project, Dubai [December, 2025 – Present]

- Leading development of a private payments system, verifiable on public blockchain, using zero-knowledge cryptography.

PhD in Applied Cryptography @ Radboud University, The Netherlands [September, 2025 – Present]

- Supervised by Prof. Joan Daemen and co-supervised by Dr. Gilles Van Assche.
- Working on design and analysis of hardware architecture-aware symmetric-key and post-quantum cryptography schemes.

Senior Cryptography Engineer @ CRC, TII, Abu Dhabi [June, 2023 – May, 2025]

- Led development, testing and release of a side-channel resistant symmetric-key and post-quantum cryptography library in `C`.

Blockchain Engineer @ Polygon Labs, Dubai [July, 2020 - May, 2023]

- Helped develop zero-knowledge virtual machine (zkVM) using STARK.
- Designed and developed peer-to-peer networked client for erasure-coded data store.
- Led smart contracts development and maintenance, secured > $7B back then.
- Designed, developed and maintained `Go` and `Rust` micro-services for event indexing and cross-blockchain asset transfer.

Open-Source :

- Developed `C++20` libraries for ML-KEM, ML-DSA, SLH-DSA, Falcon, FrodoKEM, Ascon, SHA3 and BLAKE3.
- Developed Private Information Retrieval (PIR) library in `Rust`, with GPU acceleration.
- Developed erasure-coding library in `Rust`, exploiting SIMD parallelism.
- Developed constant-time cryptography library in `C++20`.

Talks :

- Practical constant-time programming at Cedarcrypt 2026.
- Rethinking PQC implementations with modern `C++` at RWPQC 2026.
- Systematic testing methodology for cryptographic software at Gelrecrypt 2025.

Academics :

Course Name	University	Year of Passing	CGPA
M.Sc. (Computer Science)	Visva-Bharati, Santiniketan, India	2020	8.63
B.Sc. (Computer Science)	Visva-Bharati, Santiniketan, India	2018	7.81

Interest :

Design and Implementation of Accelerated Cryptographic Schemes, Privacy Preserving Computation.

Skills :

Responsibility, Patience, Diligence